

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Advanced Database**

Code No: **CSC461** Semester: **8th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Discuss the reasons for converting SQL queries into relational algebra. What is meant by the term heuristic optimization? Discuss the main heuristics that are applied during query optimization with an example.
2. List the types of transparency in distributed database. Explain the distributed database architecture.
3. How do spatial databases differ from regular databases? What are deductive databases? How do you apply trigger in active database? Illustrate with an example.

Section B

Attempt any Eight questions[8×5=40]

4. Discuss the replication and allocation techniques for distributed database design.
5. State CAP theorem. What is meant by cost based query optimization?
6. Explain the features of NOSQL system.
7. Discuss the two main types of constraints on specializations and generalizations.
8. Discuss about super type and sub type with an example.
9. Distinguish between disjointness and overlap constraints.
10. Describe about the different graphical notations for representing ODL schemas.
11. How do you map EER to relation? Illustrate with an example.
12. How do you specify object persistence via naming and reachability?